

T4.

Q5. 6-axis $45^\circ \rightarrow 90^\circ$ in 7s

$$\theta(t) = C_0 + C_1 t + C_2 t^2 + C_3 t^3$$

$$\begin{cases} \theta(0) = C_0 = 45^\circ \end{cases}$$

$$\theta(7) = C_0 + 7C_1 + 49C_2 + 343C_3 = 90^\circ$$

$$\dot{\theta}(t) = C_1 + 2C_2 t + 3C_3 t^2$$

$$\begin{cases} \dot{\theta}(0) = C_1 = 0 \end{cases}$$

$$\dot{\theta}(7) = C_1 + 14C_2 + 147C_3 = 0$$

$$\Rightarrow \begin{cases} C_0 = 45^\circ \\ C_1 = 0 \\ C_2 = 2.755^\circ \\ C_3 = -0.262 \end{cases} \quad \begin{cases} 49C_2 + 343C_3 = 45^\circ \\ 14C_2 + 147C_3 = 0 \end{cases}$$

$$\Rightarrow \theta(t) = 45 + 2.755t^2 - 0.262t^3$$

$$\dot{\theta}(t) = 5.510t - 0.786t^2$$

$$\ddot{\theta}(t) = 5.510 - 1.572t$$

Q8. $\theta(t) = C_0 + C_1 t + C_2 t^2 + C_3 t^3 + C_4 t^4$

$$\begin{cases} \theta(0) = C_0 = 10^\circ \end{cases} \quad (1)$$

$$\theta(5) = C_0 + 5C_1 + 25C_2 + 125C_3 + 625C_4 = 70^\circ \quad (2)$$

$$\dot{\theta}(t) = C_1 + 2C_2 t + 3C_3 t^2 + 4C_4 t^3$$

$$\dot{\theta}(0) = C_1 = 0 \quad (3)$$

$$\int \ddot{\theta}(t) = c_1 + 10c_2 + 75c_3 + 500c_4 = 0 \quad (4)$$

$$\dot{\theta}(4) = 2c_2 + 6c_3t + 12c_4t^2$$

$$\dot{\theta}(0) = 2c_2 = 4 \quad (5)$$

$$\Rightarrow \left\{ \begin{array}{l} c_0 = 10^0 \\ c_1 = 0 \\ c_2 = 2 \\ c_3 = 1/12 \\ c_4 = -0.208 \end{array} \right. \quad \begin{array}{l} 125c_3 + 625c_4 = 70 - 10 - 50 \\ = 10 \\ 75c_3 + 500c_4 = -20 \\ \left\{ \begin{array}{l} 25c_3 + 125c_4 = 2 \\ 15c_3 + 100c_4 = -4 \end{array} \right. \end{array}$$

$$\Rightarrow \theta(t) = 10^0 + 2t^2 + 1/12 t^3 - 0.208 t^4$$

$$\dot{\theta}(t) = \dots$$

$$\ddot{\theta}(t) = \dots$$

$$\text{Q9} \quad \theta_0 = 10^0 \quad \omega = 10^0/s \quad t_F = 10s \\ \theta_F = 45^0$$

$$\int 2 \times \frac{\dot{\theta}^2}{2\ddot{\theta}} + \dot{\theta} \cdot (t_F - 2t_b) = \theta_F - \theta_0 \quad (1)$$

$$\frac{\dot{\theta}}{\dot{\theta}} = t_b \quad (2)$$

$$\Rightarrow \begin{cases} \ddot{\theta} = \frac{12}{65} = \frac{20}{13} \\ t_b = 6.5 \text{ s} \end{cases}$$

9.999

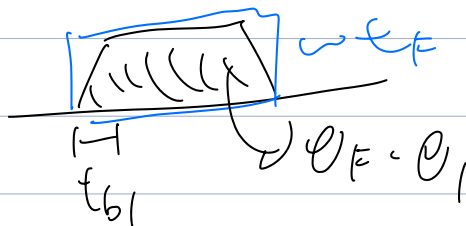
$$(10 - t_b) = 3.5^\circ$$

Q10.

Motion 1:

$$\theta_1 = 13^\circ \quad \theta_F = 37^\circ \quad \omega = 3^\circ/\text{s} \quad t_F = 9.5 \text{ s}$$

$$t_b = \frac{\omega t_F - (\theta_F - \theta_1)}{\omega} = 1.5 \text{ s}$$



$$\text{ini: } \theta(t) = 13^\circ + \frac{1}{2} \cdot 2 \cdot t^2 = 13 + t^2$$

$$\dot{\theta}(t) = 2t$$

$$\ddot{\theta}(t) = 2$$

$$\text{mid: } \theta(t) = \omega_1 (t - t_{b1}) + \theta_{A1} = 3t + 10.75$$

$$\dot{\theta}(t) = 3$$

$$\ddot{\theta}(t) = 0$$

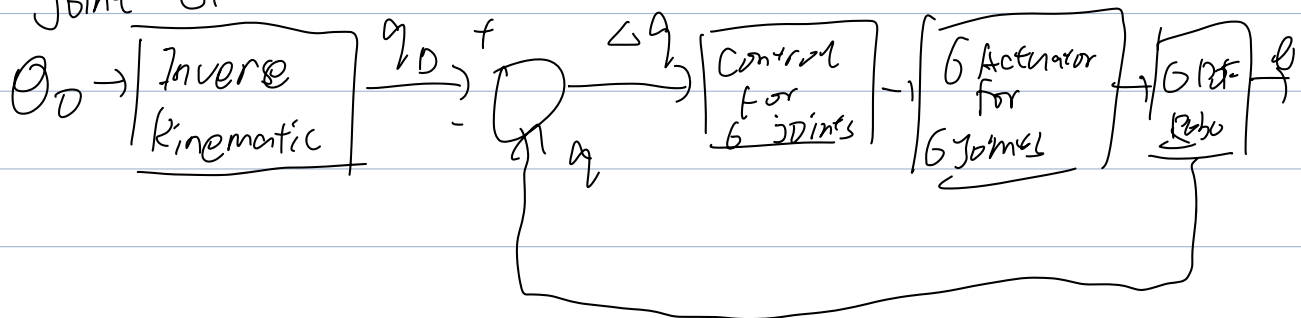
$$\begin{aligned} \text{final: } \theta(t) &= \theta_{F1} - \frac{1}{2} \cdot 2 (t_F - t)^2 \\ &= 370 - (t^2 - 19t + 90.25) \\ &= -t^2 + 19t - 53.25 \end{aligned}$$

$$\dot{\theta}(t) = -2t + 19$$

$$\ddot{\theta}(t) = -2$$

Bill

Joint Space



Operational Space

